

Fast-Charging EV Battery Thermal Management Systems

Phase 1 Research Findings Synthesis & Phase 2 Experimental Roadmap

PROJECT METADATA & INSTITUTIONAL CREDIT

Project Type: Study-Oriented Research Project (Phase 1 Synthesis)
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Domain: Fast Charging EV Battery Thermal Management Systems (BTMS)

1. Executive Summary & Project Context

Electric Vehicle (EV) adoption and consumer acceptance critically depend on achieving extreme fast charging (XFC) capabilities without compromising battery health or vehicle safety. High C-rate charging and discharging (2C to 5C) generate excessive Joule and electrochemical heat within lithium-ion battery modules. When cell temperatures exceed 50 °C, capacity degradation accelerates rapidly, cell-to-cell thermal imbalance worsens, and the risk of catastrophic thermal runaway increases dramatically. Effective Battery Thermal Management Systems (BTMS) are therefore essential to regulate peak temperatures and eliminate thermal non-uniformity across battery packs.

This study-oriented project, conducted under the guidance of Dr. Srinibas Tripathy at BITS Pilani, Rajasthan, focuses on evaluating, benchmarking, and synthesizing advanced BTMS paradigms. Phase 1 of this project comprises a comprehensive literature synthesis across five foundational computational fluid dynamics (CFD) studies. These studies encompass active casing liquid cooling, passive Phase Change Materials (PCM), finned thermal enhancement geometries, multi-hybrid systems combining PCMs, Heat Pipes (HP), Thermoelectric Coolers (TEC), and liquid cooling plates, as well as nano-enhanced PCMs (NePCM).

2. Phase 1 Synthesized Findings Across BTMS Paradigms

2.1 Novel Active Casing Liquid Cooling Architecture

Active liquid cooling remains a benchmark for high-heat-flux dissipation. A novel active casing design evaluated a 25-cell (5×5) 18650 lithium-ion battery module wherein each cylindrical cell is enclosed within an aluminum casing. The casing array is submerged directly in a square cross-flow duct through which water circulates, acting as a structural compromise between direct and indirect liquid cooling. Key empirical and numerical takeaways include:

- **Extreme Discharge Suppression:** Under an uncooled 5C discharge rate, the battery pack experiences a severe temperature rise exceeding 37.6 °C.

- **Forced Liquid Cooling Efficiency:** Introducing forced water cooling even at a minimal inlet velocity of $u = 0.01$ m/s suppresses maximum pack temperature to below 28°C under 5C discharge conditions.
- **Ultra-Low Cell Temperature Difference:** Cell-to-cell temperature variations across adjacent cells are restricted to an extraordinary 0.12°C , well below the standard industrial safety ceiling of 5°C .
- **CFD Validation Rigor:** Numerical 3D CFD models were rigorously validated against established empirical heat transfer correlations for fluid cross-flow through inline tube banks.

2.2 Passive PCM Cooling & Fin Geometry Optimization

Passive thermal management utilizing Phase Change Materials (PCM), specifically paraffin wax, leverages latent heat absorption during phase transition to absorb thermal energy without parasitic electrical power consumption. However, pure paraffin suffers from low intrinsic thermal conductivity. Integrating aluminum fins creates conductive pathways within the PCM matrix. Comparative analysis of 18650 cell cooling yields critical operational insights:

- **Working Time Extension:** An air-cooled cell reaches the critical safety threshold of 60°C in just 400 seconds. Enclosing the cell in pure paraffin PCM extends working time to 1800 seconds (a 350% increase).
- **Fin Count & Geometry Effects:** Adding 4 circular aluminum fins increases working time to 2000 seconds (+11.11%), while 8 circular fins extend working time to 2200 seconds (+10%). Increasing fin count to 12 circular fins yields no additional performance gain due to PCM displacement and thermal saturation.
- **Circular vs. Longitudinal Fins:** Longitudinal fins conduct heat more rapidly during early transient operation (< 600 s), whereas circular fins deliver superior heat dissipation throughout the main latent phase-change process.
- **Cell Pitch Spacing:** Expanding cell spacing from 8 mm to 12 mm increases the surrounding PCM volume buffer, extending safe working time up to 2600 seconds.

2.3 Advanced Multi-Hybrid Cooling Systems (PCM + HP + TEC + Liquid)

To meet extreme heat dissipation demands during fast charging, multi-hybrid BTMS architectures combine passive and active cooling mechanisms. A 12-cell module integrating Thermoelectric Coolers (TEC), Heat Pipes (HP), Paraffin Wax (PW) PCM, and liquid cooling plates was evaluated using CFD-extended multiphysics modeling. The findings demonstrate key synergistic effects:

- **Synergistic Heat Suppression:** The multi-hybrid configuration outperforms standalone and dual cooling systems, reducing peak module temperature by 92.85 K compared to PCM-only systems.
- **TEC Input Current Optimization:** The optimal input current for the Thermoelectric Cooler is established at 1.25 A. Increasing current beyond 1.5 A overcools corner cells, degrading cell-to-cell temperature uniformity across the pack.
- **Expanded Graphite (EG) Enhancement:** Incorporating 12 wt% Expanded Graphite (EG) into paraffin wax overcomes thermal resistance. Under high heat generation (6 W/cell), maximum module temperature drops to 299.2 K ($\sim 26^{\circ}\text{C}$) and cell temperature variation is held to 3.46 K.

2.4 Parametric Fin & PCM Integration Dynamics

Parametric CFD evaluations comparing bare 18650 batteries against configurations with annular fins (2, 4, 6, 8 fins), Paraffin RT-42 PCM, and hybrid fin-PCM arrangements under 2C and 3C discharge rates revealed clear comparative advantages:

- **Baseline vs. Hybrid Performance:** Uncooled batteries reach 51.6 °C at 2C and 84.8 °C at 3C. Annular fins alone reduce peak temperature by 50.5% at 3C; PCM alone achieves a 51.24% reduction; and the hybrid PCM-fin system achieves a 55.3% reduction, maintaining cell temperatures below 35 °C.
- **Optimal Structural Design:** A configuration of 4 annular fins embedded in PCM provides the optimal design balance, maximizing heat removal while minimizing mass penalty and manufacturing complexity.
- **Convective Coefficient Impact:** Increasing external convective heat transfer coefficients ($h = 20 \text{ W/m}^2\text{K}$) reduces the required critical insulation thickness around the battery housing.

2.5 Critical PCM Thickness & Nano-Enhanced PCM (NePCM)

Determining the exact material requirements for passive PCM jackets is vital for optimizing pack energy density. Evaluating Paraffin RT-42 thickness (1 mm to 7 mm) and Al₂O₃ nanoparticle concentration (0.5 wt% to 5 wt%) under 3C discharge (1200 s) at 35 °C ambient revealed:

- **Identification of Critical Thickness:** PCM thicknesses under 4 mm (1–3 mm) undergo complete liquid transition (liquid fraction $\beta = 1$), causing battery temperatures to exceed safe operational limits ($> 50 \text{ °C}$). A thickness of 4 mm is established as the Critical Thickness, limiting maximum temperature to 42.92 °C while retaining partial liquefaction ($\beta < 1$). Thicknesses exceeding 4 mm provide negligible thermal gains while unnecessarily increasing pack weight.
- **Nanoparticle Concentration Thresholds:** Adding 0.5 to 2 wt% Al₂O₃ nanoparticles enhances thermal conductivity, capping maximum battery temperature at 42.77 °C. However, excessive nanoparticle loading (5 wt%) degrades performance (max temperature increases to 42.88 °C) because surplus nanoparticles displace latent heat storage capacity.

3. Comparative Performance & Trade-off Matrix

To inform Phase 2 system selection, the operational trade-offs, peak thermal limits, temperature uniformity, and design complexity across all five evaluated BTMS paradigms are synthesized in the matrix below.

BTMS Strategy	Peak Temp (3C–5C)	Temp Uniformity (ΔT)	Working Duration	Primary Design Trade-off
Active Casing Liquid Cooling	$< 28.0 \text{ °C}$ (at 5C)	0.12 °C (Ultra-high)	Continuous active	High cooling efficiency; requires active pumps, piping, and mass penalty.
Pure Paraffin PCM Jacket	$> 50.0 \text{ °C}$ (Melted)	3.0 – 6.0 °C	1800 s (at 60 °C)	Zero parasitic power; limited

				by low thermal conductivity and finite latent heat.
Finned PCM (4 Annular Fins)	< 35.0 °C (at 3C)	2.0 – 4.0 °C	2000 s – 2200 s	55.3% temp reduction over bare cell; optimal balance of fin mass & PCM volume.
4 mm NePCM (1-2 wt% Al ₂ O ₃)	42.77 °C (at 3C)	1.5 – 3.0 °C	Sustained 1200 s+	Critical thickness minimizes mass; excess loading (>2 wt%) reduces latent heat.
Multi-Hybrid (PCM+HP+TEC+Liquid)	26.0 °C (299.2 K)	3.46 K (at 6 W/cell)	Continuous hybrid	Highest thermal performance (-92.85 K); complex controls and high cost.

4. Phase 2 Roadmap & Future Work Directions

Building upon the numerical benchmarks established in Phase 1, the second phase of this research under Dr. Srinibas Tripathy at BITS Pilani will transition from literature synthesis and numerical modeling toward experimental prototyping and multi-objective optimization. The key research vectors for Phase 2 include:

- **Experimental Test Bench Construction & Validation:** Fabricate physical single-cell and 2×2 module prototypes featuring the optimized 4 mm NePCM (with 1 wt% Al₂O₃ / EG) combined with 4 annular aluminum fins. Subject the prototype to fast-charging current profiles (2C, 3C, and 5C) under temperature-controlled environmental chamber conditions to validate CFD numerical predictions.
- **Full Pack-Level Scale-Up & Fluid Manifold Dynamics:** Scale 3D numerical models from module level to full pack architectures. Analyze coolant manifold flow distribution, hydraulic pressure drops, and inter-cell thermal propagation during localized thermal runaway events.
- **Multi-Objective Optimization Framework:** Formulate multi-objective Pareto optimization models using genetic algorithms (e.g., NSGA-II) to simultaneously optimize peak thermal reduction, overall pack weight and volumetric energy density penalties, manufacturing cost, and parasitic cooling power.
- **Extreme Fast Charging (XFC) & Harsh Environmental Stress Testing:** Evaluate system resilience under non-uniform dynamic fast-charging drive cycles (US06 / WLTP profiles) at elevated ambient temperatures (> 40 °C) to establish operational safety boundaries.

5. References & Source Benchmark Studies

- Study 1: Novel Active Casing Liquid Cooling for Lithium-Ion Battery Modules (01_BTMS_CFD_Reference_1.pdf).
- Study 2: Passive Cooling of Cylindrical Batteries Using Paraffin PCM with Circular and Longitudinal Fins (02_BTMS_CFD_Reference_2.pdf).

- Study 3: Multi-Hybrid Thermal Management System Integrating PCM, Heat Pipes, Thermoelectric Coolers, and Liquid Plates (03_BTMS_CFD_Reference_3.pdf).
- Study 4: Parametric CFD Evaluation and Validation of Annular Fin and PCM Integration for Cylindrical Batteries (04_BTMS_CFD_Reference_4_Validation.pdf).
- Study 5: Determination of Critical PCM Thickness and Nano-Enhanced Paraffin (Al₂O₃) Thermal Management Systems (BTMS_Critical_Thickness (1).pdf).